

EE 340 Electromagnetic Theory

Assignment #01

✓ Q.1 Calculate the gradient of the following scalar field

$V(x, y, z) = x + 2y + 3z$, at the points of coordinates $P(1, 2, 3)$ and $Q(4, -2, 9)$.

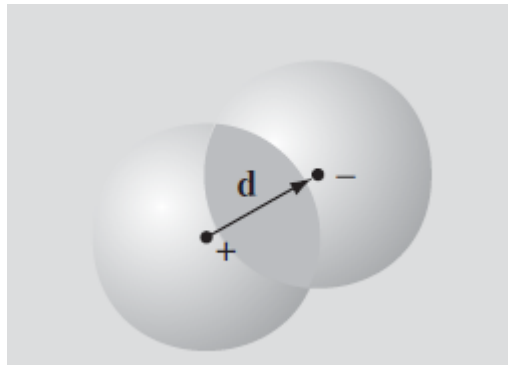
Q.2 Calculate the flux of the vector field $\vec{F} = x^2 \hat{a}_x + xy \hat{a}_y + z \hat{a}_z$ across the plane $2x + 3y + z = 6$

✓ Q.3 A thick spherical shell carries charge density

$$\rho = \frac{k}{r^2} \quad (a \leq r \leq b)$$

Find the electric field in the following three regions: (i) $r < a$, (ii) $a < r < b$, (iii) $r > b$. Plot $|E|$ as a function of r , for the case $b = 2a$.

✓ Q. 4 Two spheres, each of radius R and carrying uniform volume charge densities $+\rho$ and $-\rho$, respectively, are placed so that they partially overlap as shown in figure. Call the vector from the positive center to the negative center d . Show that the field in the region of overlap is constant, and find its value.



✓ Q.5 One of these is an impossible electrostatic field. Which one?

(a) $E = k[xy \hat{x} + 2yz \hat{y} + 3xz \hat{z}]$.

(b) $E = k[y^2 \hat{x} + (2xy + z^2) \hat{y} + 2yz \hat{z}]$.

Here k is a (nonzero) constant with the appropriate units.

✓ Q. 6 (a) Three charges are situated at the corners of a square (side length a), as shown in Figure below. How much work does it take to bring in another charge, $+q$, from far away and place it in the fourth corner?

(b) How much work does it take to assemble the entire configuration of four charges?

